

ANDREW BOND

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Education

EPFL Lausanne, Switzerland
EDIC Fellowship Student

September 2026 – Present

Koc University Istanbul, Türkiye

2023 – August 2026

Masters: Computer Science and Engineering Advisor: Assoc. Prof. Aykut Erdem

CGPA: 3.73/4.0

- Thesis Title: Matching Geometry to Data: A Ladder of Representation Learning for the 4D Visual World
- Awarded Koç Presidential Fellowship as a top graduate student/researcher (2025 – 2026 Academic Year), and AI Fellowship for top AI graduate students (2024 – 2026 Academic Years).
- Achieved two first-author publications in top-tier conferences (NeurIPS 2024, ICCV 2025), and multiple other first-author publications under review at top-tier conferences (NeurIPS 2026).
- Awarded the Academic Excellence Award for best graduating student in the department.
- Nominated for best thesis award.

Koc University Istanbul, Türkiye

2019 – 2023

Bachelors in Computer Engineering, Minor in Mathematics

CGPA: 3.55/4.0

- Awarded Vehbi Koç Scholars for top-performing academic performance in a semester (Spring 2020, Fall 2021, Spring 2021, Spring 2022).
- Awarded Dean's Honor Roll for high academic achievement (Fall 2022, Spring 2023).
- Achieved two publications in top-tier conferences and findings (ICCV2023, IJCNLP-AAACL2023)

Publications

M. H. Ali*, **Andrew Bond***, T. Birdal, D. Ceylan, L. Karacan, E. Erdem, and A. Erdem. VidStyleODE: Disentangled video editing via StyleGAN and NeuralODEs. *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 7489–7500, 2023. In collaboration with Adobe London (Duygu Ceylan) and Imperial College London (Tolga Birdal), accepted to ICCV 2023.

A. Kara, F. M. Sofian*, **Andrew Bond***, and G. G. Şahin. GECTurk: Grammatical error correction and detection dataset for turkish. In *International Joint Conference on Natural Language Processing*, 2023.

Andrew Bond and Z. Doğan. Exploring the precise dynamics of single-layer GAN models: Leveraging multi-feature discriminators for high-dimensional subspace learning. *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 2024.

Andrew Bond and Z. Doğan. Effective covariance dynamics in solvable high-dimensional gans. 2026. Under review at ACML 2026 Journal Track.

Andrew Bond, S. Demir, and Z. Doğan. High-dimensional dynamics of mixture-of-experts: Phase transitions, routing, and specialization. a. Under review at NeurIPS 2026.

Andrew Bond, E. E. Özlü, T. Çimen, I. U. Melanlioglu, T. Birdal, E. Erdem, and A. Erdem. Task-induced riemannian metrics for vision transformer feature spaces. b. Under review at NeurIPS 2026. Done in collaboration with Imperial College London (Tolga Birdal).

Andrew Bond, J.-H. Wang, L. Mai, E. Erdem, and A. Erdem. GaussianVideo: Efficient video representation via hierarchical gaussian splatting. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 7187–7196, October 2025. Done while an intern at Adobe Research.

Andrew Bond, I. U. Melanlioglu, E. Erdem, and A. Erdem. Beyond gaussian bottlenecks: Topologically aligned encoding of vision-transformer feature spaces. *arXiv preprint arXiv:2604.28122*, 2026. Under review at NeurIPS 2026, reduced version accepted at ECCV 2026 Beyond Euclidean Workshop.

H. Çapuk*, **Andrew Bond***, G. Çeliker, M. B. Kızıl, E. Erdem, and A. Erdem. Geometry-aligned tangent-plane diffusion transformers for omnidirectional image generation. *arXiv*, 2025.

^{1*} indicates equal contribution.

Internship Experience

Adobe, Video AI Lab *Research Scientist Intern*

June 2024 – September 2024

↔ Supervisor: Jui-hsien Wang

Adobe Research (Seattle, USA)

- Designed a hierarchical Gaussian splatting method for dynamic videos; resulting in a first-author ICCV 2025 paper and a patent filing.
- Can be found on ArXiv at the following link: <http://arxiv.org/abs/2501.04782>.

Patents

Digital Video Modifying Using a Gaussian Splat Patent Application Number: P13705-US, Filing Date: May 29th, 2025, Patent Pending. *Produced while an intern for Adobe Research.*

Active Research Projects

These research projects are from my time at Koç University during my master's degree, and will be finished soon. Some are with me as the primary researcher, while others I am simply a contributor or advisor on.

Generative Models

- * Working on a diffusion architecture allowing for converting arbitrary transformer architectures into generative models, with additional long-sequence generation capabilities. Plan to submit to ICLR 2027.
- * Exploring how to make generative metrics more robust to irrelevant changes in the data output (metric invariance), as well as how to develop metrics for modalities which don't easily have an embedding space to apply traditional metrics in. Done in collaboration with Imperial College London (Tolga Birdal), and plan to submit to ICLR 2027.

3D Computer Vision

- * Developing a method for using visual geometry pretrained models for feedforward 4D Gaussian Splatting. Aiming to submit to CVPR 2027.
- Trying to incorporate reflection understanding into image generative models by incorporating 3d priors into the training and/or inference. Aiming to submit to ICLR 2027 or CVPR 2027.

Visual-Audio Representations

- Exploring the possibility of a joint visual-audio version of Gaussian Splatting, where the gaussians represent both visual and audio information, and the method allows for novel synthesis of both vision and audio.

Theoretical Deep Learning

- * Working on a Dynamical Mean-Field Theory (DMFT) of Mixture-of-Experts (MoE) models in the high-dimensional proportional limit, seeking to understand the conditions under which specialization vs router collapse happens, when load-balancing losses are needed, and how router learning relates to the Random Energy Model (REM) from disordered systems theory. Under submission at NeurIPS 2026.
- * About to submit an IEEE SPL paper on studying the effects of missing data on the training of certain solvable GAN models.

²* indicates that I am the primary researcher for the project.

Teaching Experience

Koc University *Teaching Assistant*

Sept 2021 – June 2026

- **COMP 549: Deep Unsupervised Learning** (Spring 2025, Spring 2026)
Led weekly tutorial sessions for 30+ students, clarifying complex theoretical concepts and demonstrating their Python implementation. Designed Python-based homework assignments focusing on generative models.
- **COMP 541: Deep Learning** (Fall 2024, Fall 2025)
Conducted weekly lab sessions for 40+ students, assisting with PyTorch implementations of deep learning architectures. Held office hours to support students with coursework and projects.
- **COMP 132: Advanced Programming** (Fall 2023, Spring 2024)
Ran weekly programming labs for 100-150 students per semester, teaching Java programming concepts. Graded assignments and course projects.
- **MATH 204: Differential Equations** Fall 2021, Fall 2022, Spring 2023
- **MATH 203: Multivariable Calculus** Spring 2022
- Awarded "Outstanding TA Award" (Spring 2024) for exceptional student reviews and performance.

Selected Course Work

Computer Science	Mathematics
Deep Unsupervised Learning	Real Analysis
Natural Language Processing	Measure Theory + Functional Analysis
Computational Imaging	Topology + Cohomology on Manifolds
Intelligent User Interfaces	Abstract Algebra 1 & 2
Nonconvex Optimization for Machine Learning	Advanced Stochastic Processes
Reinforcement Learning	Operator Theory / Intro to PDEs
Computer Graphics	Financial Mathematics
Modern Learning Theory	

Technical Skills

Programming Languages	Python, CUDA, Metal, C++, Julia, Matlab
Languages	English (Native), Turkish (Basic)

Reviewer for Conferences

- Neural Information Processing Systems (NeurIPS), 2025, 2026
- IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2025, 2026
- IEEE Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2026
- International Conference on Machine Learning (ICML), 2026 (Gold Reviewer)